

DIFFA, Niger

WMO: 610850

Lat: 13.373N

Lon: 12.627E

Elev: 303

StdP: 97.74

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.5	13.9	-7.8	2.0	24.5	-6.7	2.2	25.0	11.5	19.7	9.9	20.7	0.9	40	0.398

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	14.7	43.0	21.8	42.1	21.7	41.1	21.8	27.0	33.5	26.6	33.1	26.2	32.8	2.6	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.5	21.5	29.8	25.0	20.9	29.3	24.6	20.4	29.1	86.8	33.2	84.8	33.2	83.1	33.0	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.0	7.9	7.0	10.2	44.5	1.0	0.3	9.4	44.8	8.8	44.9	8.2	45.1	7.5	45.3
			4.8	28.0	0.5	0.3	4.4	28.2	4.1	28.4	3.8	28.5	3.4	28.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.9	22.4	25.4	29.4	33.2	34.4	33.2	30.3	28.1	29.8	30.6	27.2	22.9
	DBStd	4.32	2.58	3.08	2.37	2.10	1.74	2.25	2.24	1.79	1.74	1.40	2.01	2.22
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	1	1	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6903	383	431	602	696	755	696	631	562	595	637	514	400
	CDD18.3	3862	126	198	344	446	497	446	372	304	345	379	264	142
	CDH23.3	51164	1262	2262	4641	6823	7725	6666	4910	3302	4186	4938	3080	1369
	CDH26.7	30576	548	1200	2873	4668	5354	4339	2678	1433	2203	2955	1748	577
(14) Wind	WSAvg	2.3	2.6	2.5	2.2	1.8	2.2	2.8	2.8	1.9	1.8	1.7	2.2	2.7
(15) Precipitation	PrecAvg	288	0	0	0	2	6	25	90	120	42	4	0	0
	PrecMax	479	0	4	5	12	46	95	202	218	97	25	1	0
	PrecMin	179	0	0	0	0	0	2	35	40	3	0	0	0
	PrecStd	85	0	1	1	3	10	26	48	40	24	8	0	0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.7	39.8	42.3	44.0	44.2	42.7	40.1	36.5	39.8	40.6	38.8	36.4
		MCWB	18.1	19.6	20.7	21.5	22.4	23.4	24.3	25.9	23.5	21.4	19.1	18.9
	2%	DB	34.4	37.8	40.7	43.1	43.2	41.4	38.2	34.9	38.1	39.6	37.5	33.7
		MCWB	16.6	17.9	19.9	21.1	22.2	23.7	24.5	26.1	24.4	21.5	19.2	17.0
	5%	DB	32.2	35.6	39.1	42.1	42.3	40.2	36.7	33.7	36.8	38.5	36.2	32.0
		MCWB	15.5	17.0	18.6	20.7	22.2	23.8	24.8	25.9	24.9	21.6	18.7	16.1
	10%	DB	29.9	33.5	37.4	40.8	41.2	38.8	35.3	32.6	35.2	37.3	34.7	30.1
		MCWB	14.6	16.1	18.1	20.5	22.2	24.1	24.8	25.8	25.3	21.5	18.1	15.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.0	21.3	24.6	24.9	26.6	26.6	27.1	27.6	27.6	26.7	21.4	19.4
		MCDB	35.1	36.9	37.3	39.7	39.5	36.2	34.4	33.2	34.0	34.0	35.3	34.8
	2%	WB	17.2	18.7	21.8	24.0	25.6	26.0	26.5	27.0	26.9	25.9	20.3	17.9
		MCDB	32.7	35.5	37.3	38.3	38.0	36.0	33.6	32.4	33.3	33.3	34.9	32.1
	5%	WB	15.9	17.6	20.1	23.2	25.0	25.6	26.0	26.5	26.4	25.2	19.3	16.7
		MCDB	31.0	34.3	36.2	37.1	37.4	35.4	32.9	31.7	32.8	32.7	34.2	30.4
	10%	WB	14.9	16.5	18.7	22.2	24.4	25.1	25.6	26.1	26.0	24.5	18.5	15.8
		MCDB	29.2	32.4	35.8	36.6	36.5	34.7	32.3	31.1	32.4	31.9	33.4	29.1
(35) Mean Daily Temperature Range	5% DB	MDBR	15.9	16.3	16.5	16.1	14.7	11.5	9.7	8.6	11.1	14.6	16.3	15.5
		MCDBR	18.8	18.4	18.9	17.4	16.0	13.1	11.3	10.2	13.2	16.4	17.5	17.8
		MCWBR	8.5	7.7	7.7	6.6	6.2	4.0	3.0	3.2	3.8	5.3	7.3	8.1
	5% WB	MCDBR	18.4	17.8	16.9	14.8	13.3	11.3	9.8	9.4	10.9	12.2	16.5	16.7
		MCWBR	8.8	7.9	7.8	6.3	4.9	3.7	3.2	3.7	3.9	4.4	7.7	8.1
(40) Clear-Sky Solar Irradiance	taub		0.555	0.618	0.665	0.745	0.725	0.707	0.674	0.624	0.628	0.660	0.579	0.574
	taud		1.701	1.571	1.493	1.400	1.451	1.535	1.650	1.757	1.729	1.595	1.655	1.665
	Ebn at Noon		743	717	697	643	644	647	672	716	715	677	716	711
	Edh at Noon		234	277	305	334	312	283	253	230	237	266	243	237
(44) All-Sky Solar Radiation	RadAvg		5.71	6.15	6.57	6.72	6.67	6.33	5.89	5.64	6.12	6.08	5.86	5.52
	RadStd		0.14	0.14	0.29	0.23	0.28	0.27	0.17	0.26	0.13	0.18	0.16	0.14

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page